Billionaire Age Distribution by Gender and Country

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1 Introduction

Wealth is unevenly distributed across populations, and demographic comparisons can help describe patterns among billionaire rankings. This project investigates the research question: *How do the age distributions of male and female billionaires compare, and does this vary across countries?*. The objective is to quantify differences in age by gender and to examine whether the same pattern occurs in the countries with the largest numbers of billionaires in the dataset.

2 Dataset and Preparation

The public Billionaires Statistics Dataset (2023) contains 2,640 billionaire records, of which 2,203 are male and 337 are female [1]. It includes 35 variables, including age, gender. country of citizenship, net worth and industry. The present analysis uses age, gender and country of citizenship. Python's pandas library was used to load the CSV file [2]. For the overall age comparison, rows missing age or gender were removed using dopna(); for the country analysis, rows missing any of the three selected variables were excluded. The country comparison was restricted to five countries with the most billionaire records after this cleaning.

3 Methods

Analysis was performed in Python using pandas, NumPy, Matplotlib and SciPy [3]. Male and female ages were summarised using the mean, median and sample standard deviation. Overlaid, density-normalised histograms used common five-year age bins; empirical cumulative distributions were used to compare the percentage of individuals below each age. Skewness and excess kurtosis were calculated to describe the shape. For each country-gender group, the mean age and its standard error were calculated:

$$SE(\bar{x}) = \frac{s}{\sqrt{n}} \tag{1}$$

where s is the sample standard deviation and n is the number of observations in that group. Error bars on the country plot show ± 1 standard error (SE). The cumulative plots use approximate pointwise uncertainty of

$$\sqrt{\frac{p(1-p)}{n}}$$

for an estimated proportion p.

4 Results and Discussion

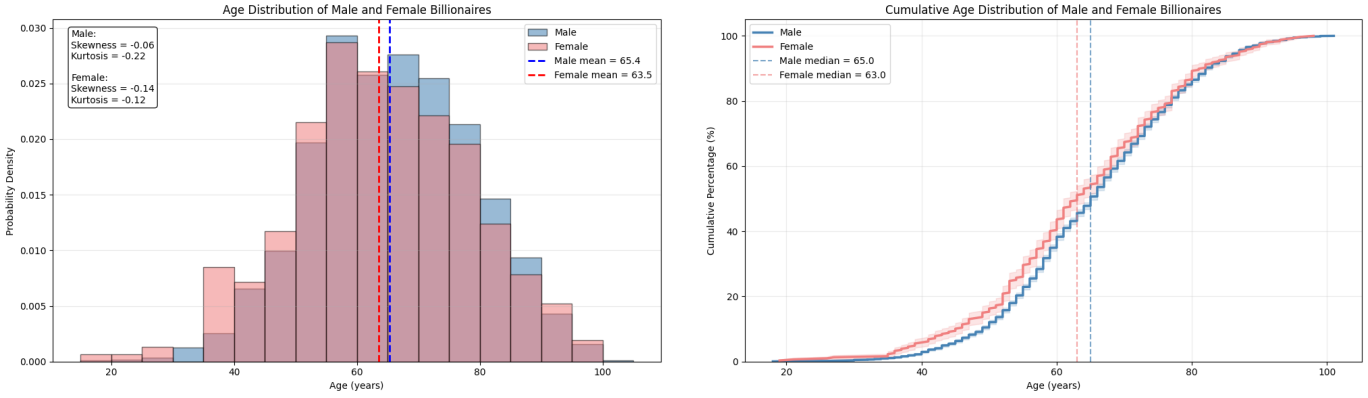


Figure 1: Age distributions of male and female billionaires: (left) histogram and (right) cumulative age distribution with uncertainty bands.

Male billionaires had a mean age of 65.4 years, compared with 63.5 years for females, a difference of 1.9 years. The corresponding medians were 65 and 63 years. Although females were slightly younger, the substantial overlap between the distributions (Figure 1) indicates broadly similar age patterns.

Skewness values (male: -0.06, female: -0.14) were close to zero, indicating approximately symmetric distributions. Excess kurtosis values (male: -0.22, female: -0.12) were also near zero, suggesting tail behaviour similar to a normal distribution. The cumulative distributions further supported these findings, with considerable overlap and slightly younger female ages.

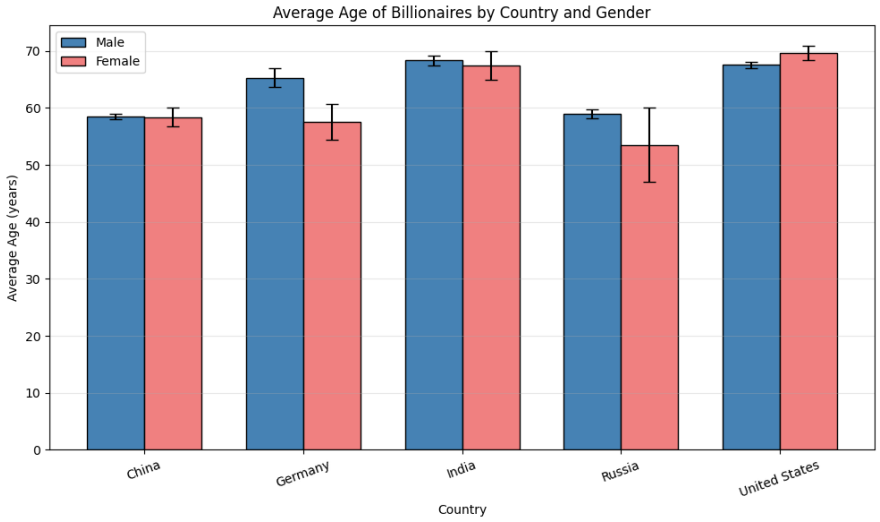


Figure 2: Mean age of male and female billionaires across the five countries with the largest billionaire populations. Error bars represent ±1 standard error of the mean.

Figure 2 shows that mean billionaire ages varied across the five countries with the largest billionaire populations, with no consistent gender pattern. Error bars represent ±1 standard error of the mean, indicating the precision of each estimate. Large error bars, particularly among some female groups, suggest greater uncertainty due to smaller samples or higher age variability. Therefore, the observed differences should not be interpreted as definitive, as further statistical analysis is needed to establish their significance, and the findings cannot demonstrate causation.

5 Conclusion

The analysis found that male and female billionaires have broadly similar age distributions, with females being slightly younger on average. The substantial overlap between distributions and skewness and kurtosis values close to zero suggest that gender is not strongly associated with age in this dataset. Although average age varied across the five countries, no consistent gender pattern was observed. Overall, age distributions were relatively similar across genders, while geographical difference were more variable.

References

[1] Guemroud, L. (2023). *Billionaires Statistics Dataset (2023)*. Kaggle. https://www.kaggle.com/datasets/lounesguemroud/billion statistics-dataset-2023

[2] McKinney, W. (2010). Data structures for statistical computing in Python. *Proceedings of the 9th Python in Science Conference*, 56–61.

[3] Virtanen, P., et al. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17, 261–272. https://doi.org/10.1038/s41592-019-0686-2